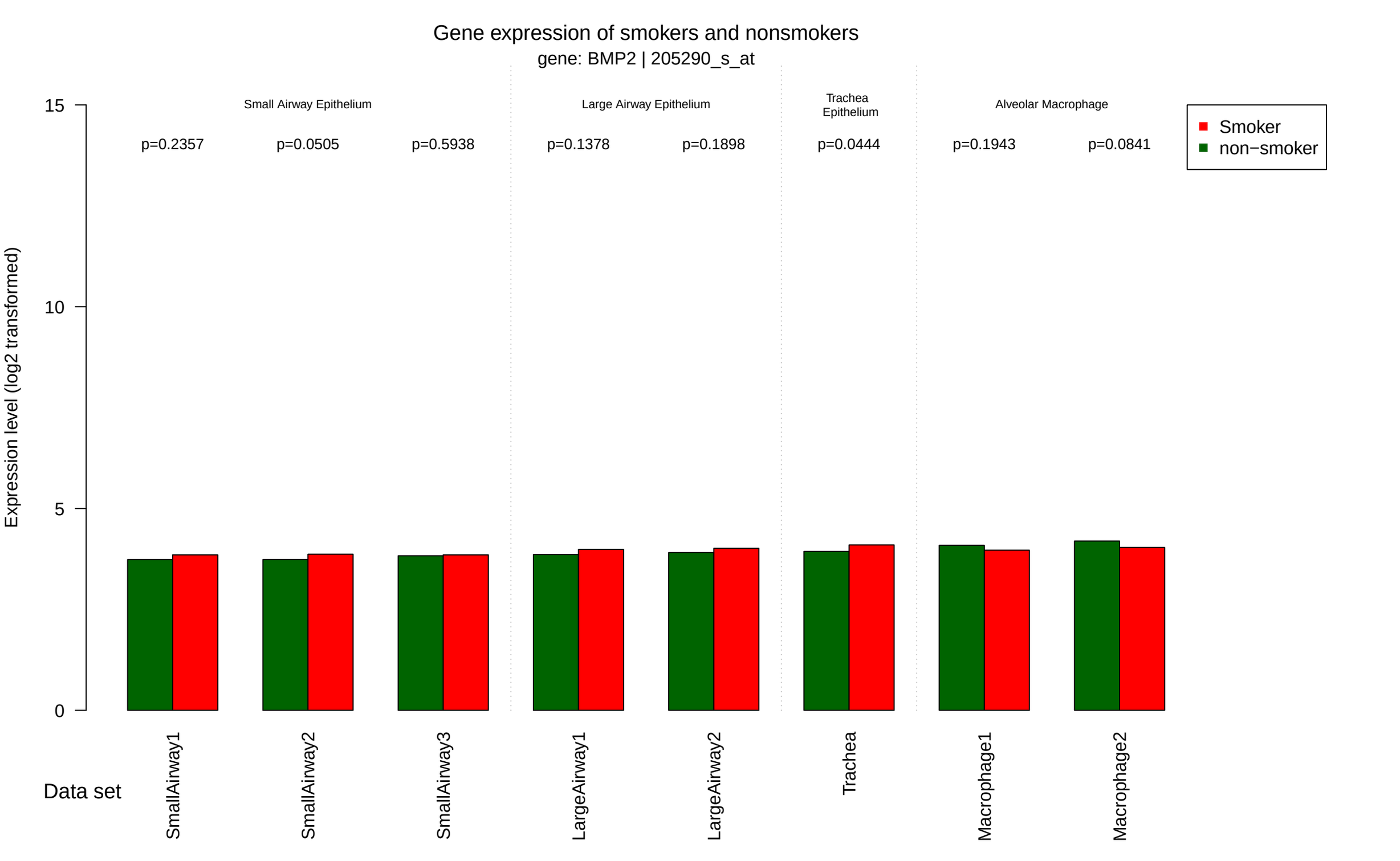
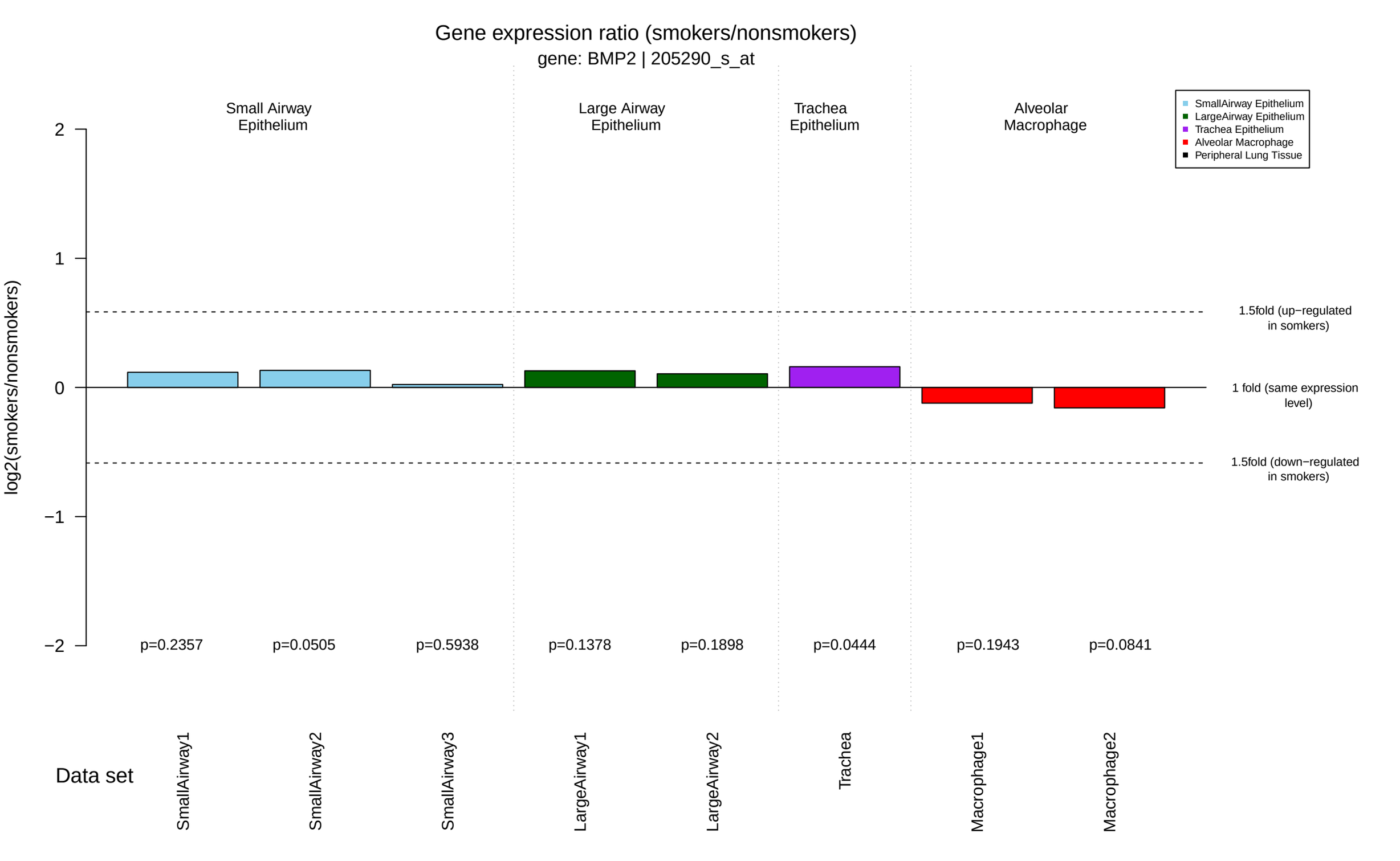
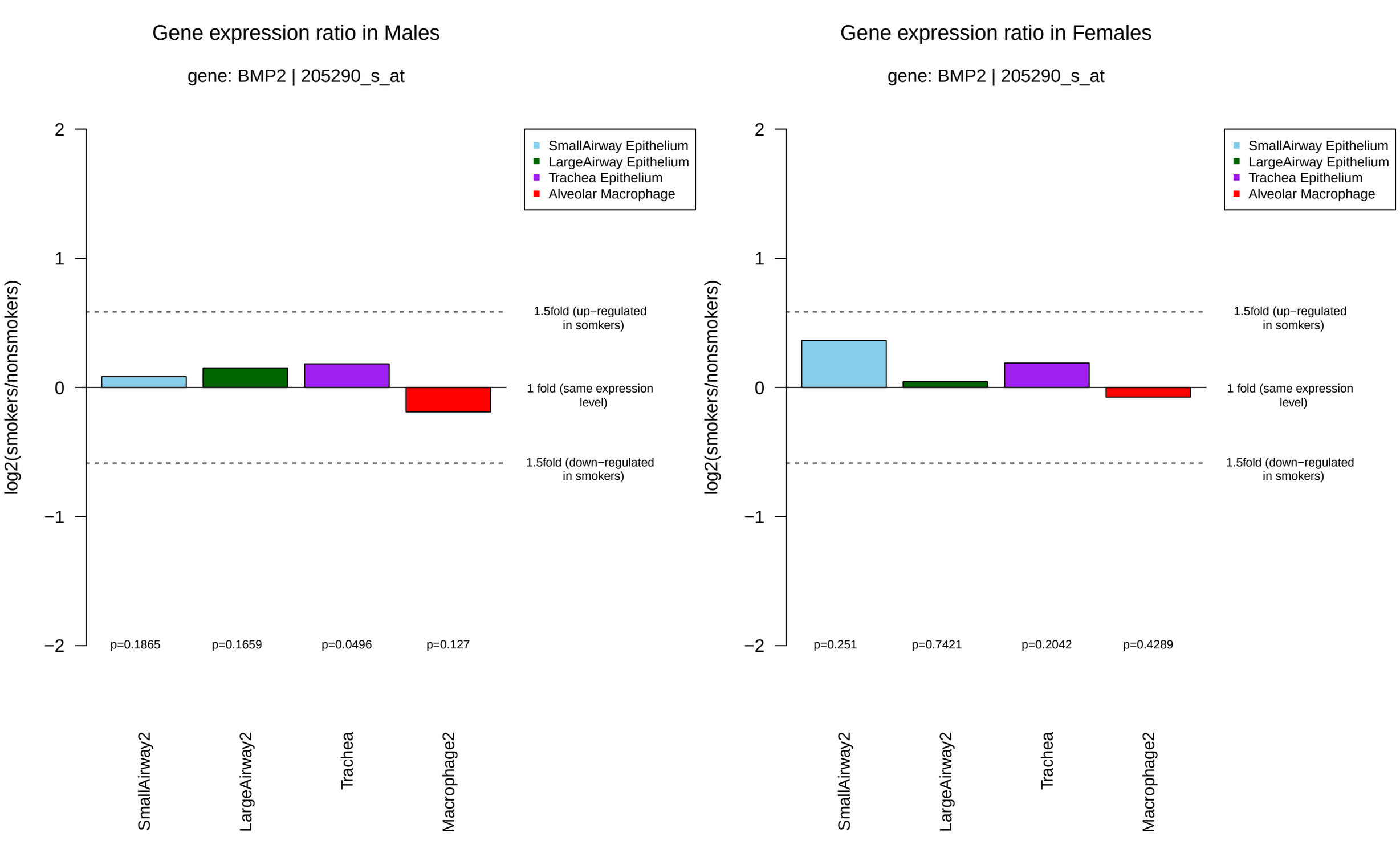
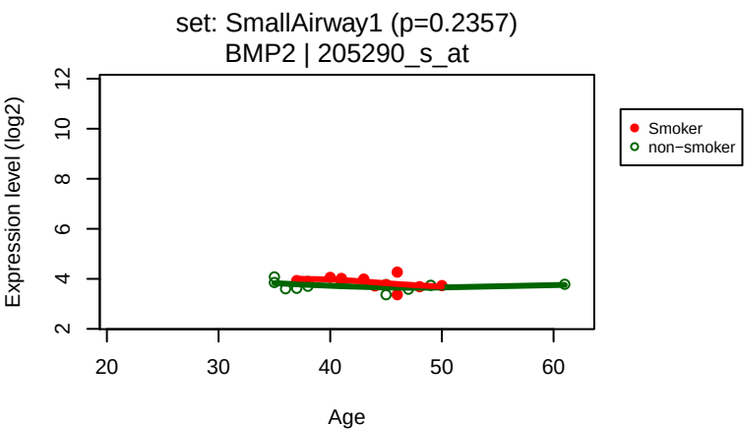
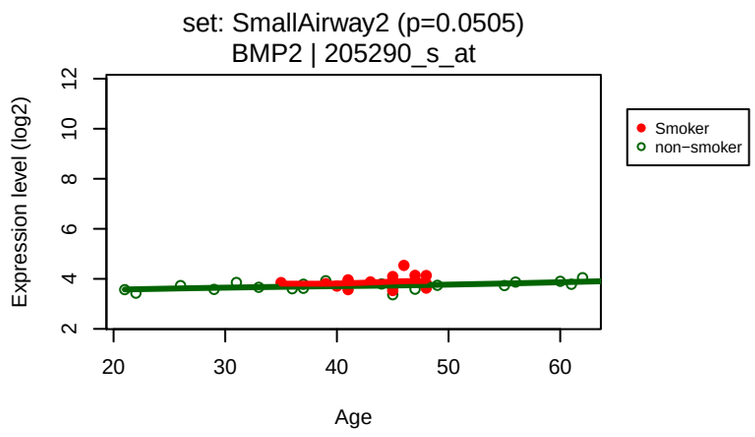
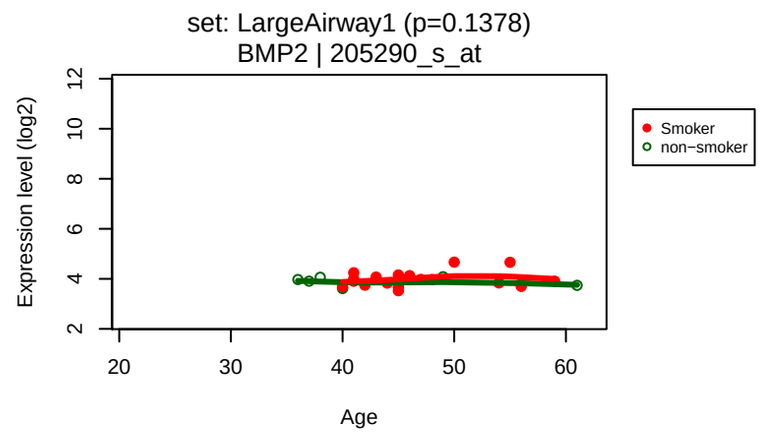

Gene expression of smokers and nonsmokers
gene: BMP2 | 205290_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2357
p=0.0505
p=0.5938
p=0.1378
p=0.1898
p=0.0444
p=0.1943
p=0.0841
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

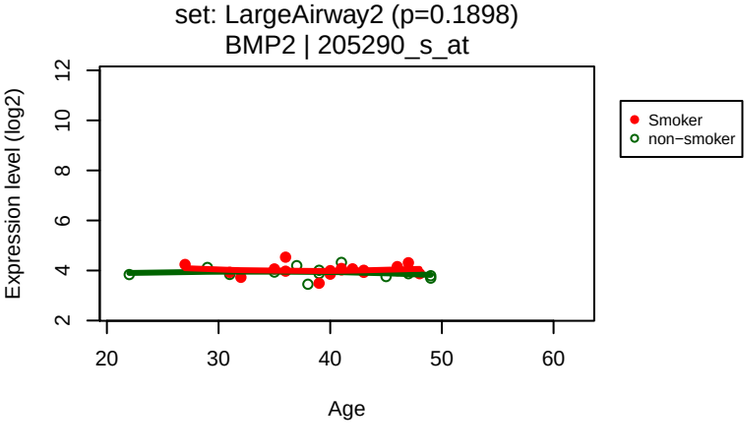

Gene expression ratio (smokers/nonsmokers)
gene: BMP2 | 205290_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2357
p=0.0505
p=0.5938
p=0.1378
p=0.1898
p=0.0444
p=0.1943
p=0.0841
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

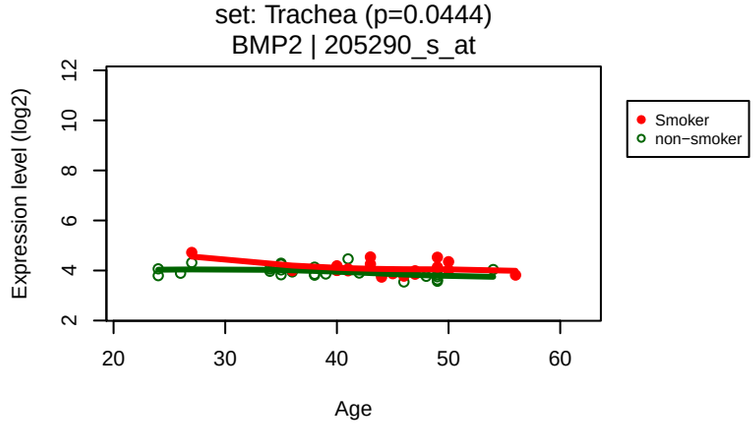

Gene expression ratio in Males
gene: BMP2 | 205290_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1865
p=0.1659
p=0.0496
p=0.127
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BMP2 | 205290_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.251
p=0.7421
p=0.2042
p=0.4289
SmallAirway2
LargeAirway2
Trachea
Macrophage2

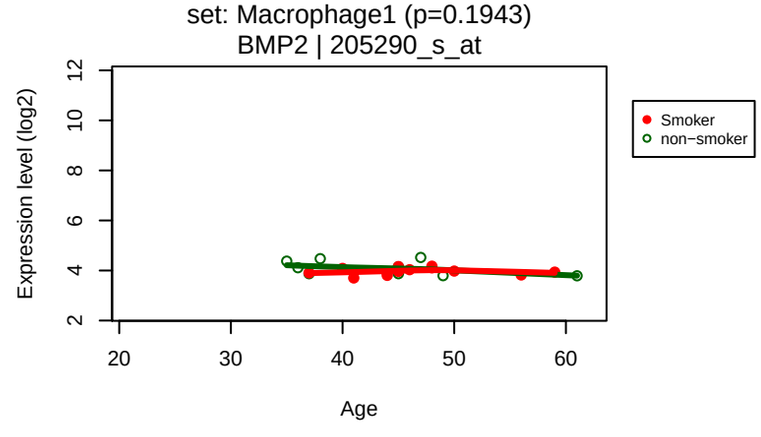

set: SmallAirway1 (p=0.2357)
BMP2 | 205290_s_at
Expression level (log2)
Age
Smoker
non−smoker

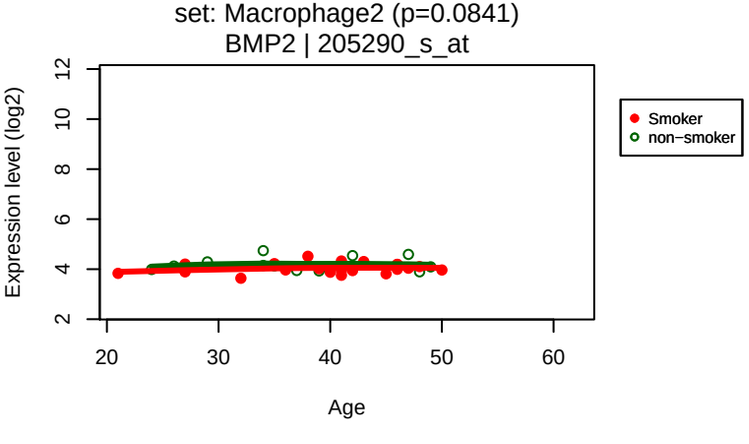

set: SmallAirway2 (p=0.0505)
BMP2 | 205290_s_at
Expression level (log2)
Age
Smoker
non−smoker

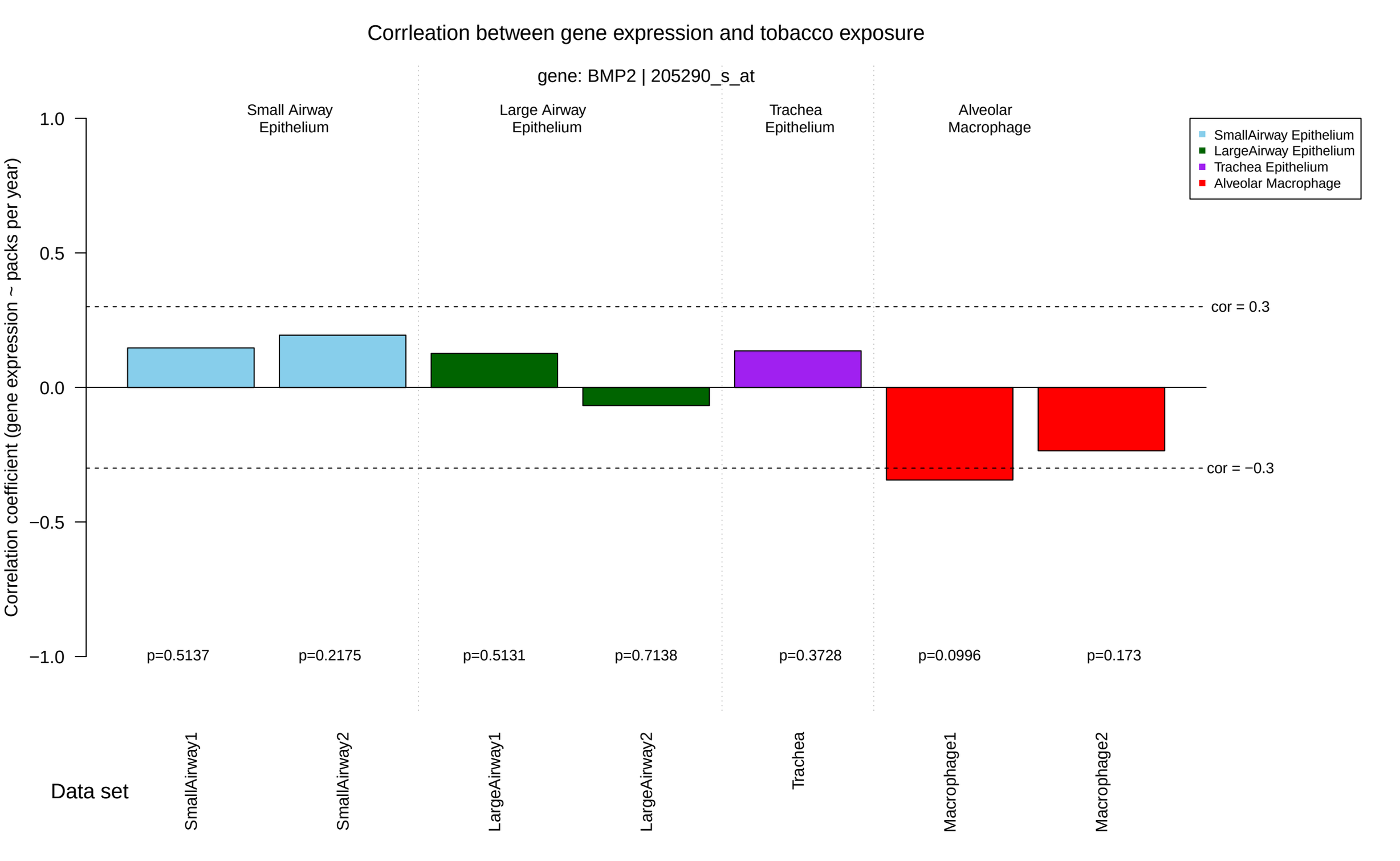

set: LargeAirway1 (p=0.1378)
BMP2 | 205290_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1898)
BMP2 | 205290_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0444)
BMP2 | 205290_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1943)
BMP2 | 205290_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0841)
BMP2 | 205290_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BMP2 | 205290_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5137
p=0.2175
p=0.5131
p=0.7138
p=0.3728
p=0.0996
p=0.173
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2